

important for an LCD. Unlike CRTs, in which these depend on the phosphor screen, LCDs have a fluorescent lamp built into the LCD monitor's casing. The brighter the lamp, the higher the luminance, and the better the monitor. The lamp rating is what determines the brightness level the monitor can achieve. A higher luminance means the LCD is easier to view in a bright room and from a distance.

In the 15-inch category, the NU L511E had the highest luminance rating at 300 cd/m² (candelas per square metre). The rest of the 15-inch category had a luminance of 250 cd/m². Under the 17-inch category, the Asus PM17TU had a luminance rating of 500 cd/m²—the highest we've seen amongst all the LCDs featured in this test. Note that this is a vendor-specified rating, and a rating of 500 means that the LCD can look bright even in a fully-lit room. The CMV CT-712A, with a luminance of 400 cd/m², had the second-highest luminance rating. We were surprised to see that the highest luminance in the 19-inch category was 330 cd/m² of the CMV CT-937A,

while most of the LCDs in this category had luminance ratings of 300 or less. The lowest luminance rating was seen in the 19-inch Philips 190V—240 cd/m².

Contrast Ratio

The ratio of the brightness of pure white versus pitch black that the monitor is capable of displaying is known as its contrast ratio. Monitors with a higher contrast ratio display images with better visual depth. A monitor with a higher

contrast ratio also allows you to fine-tune its contrast settings to suit different lighting conditions.

As a rule of thumb, a contrast ratio of 120:1 is enough to display most colours. When the contrast ratio is even higher, the LCD is able to display more levels of grey. What this means is that with an LCD with a high contrast ratio, you will be able to see more shades of colour.

The NU L511E and the Samsung SyncMaster 540N claimed the highest contrast ratio of 500:1 amongst the 15-inchers. The others had contrast ratios of 450:1 and 400:1. In the 17-inch LCDs, the BenQ FP71E+ boasted of a very high contrast ratio of 1000:1—the highest of all LCDs in the test. This means the user has a wide range of contrast and brightness to play with. Most others in this category had a contrast ratio of 600:1 or 500:1. The ViewSonic VA712 had the lowest contrast ratio of 350:1, leaving little room for tweaking.

In the 19-inch models, the highest contrast ratio—800:1—was seen on the NEC MultiSync LCD1920NXp, followed by the Acer AL1916W with 700:1.



Samsung
SyncMaster 540N

How We Tested

The monitors were tested keeping in mind performance, features, and value for money. Before running the tests, the monitors were placed perfectly vertically in order to eliminate any uniformity issues, and were auto-calibrated. All the monitors were set to their native resolutions, and the drivers were loaded for best performance.

The Test Bed: Both the reference and the test systems were running on an Intel Pentium 4 3.0 GHz with 1 GB of DDR SDRAM, and were powered by an nVidia 5950 Ultra graphics card. The test rigs had a 120 GB 7200 rpm SATA hard drive. Windows XP Professional SP1 was loaded, along with the latest version of DirectX and other device drivers.

Features: In the Features tests, we looked for value-additions and other aspects that increased the user-friendliness of a display. The features noted varied in significance—some were simple, like the user-friendliness of the OSD (On-Screen Display); some were important, such as power consumption, depth and weight. One important thing we looked at was the monitor base, which decides the tilt the monitor is capable of—the more the better. Some LCDs can be rotated vertically for both landscape and portrait orientations. A portrait orientation is especially useful while reading a long Word file or a PDF document, as it reduces the need to scroll off a screen. We also rated additional features such as Web camera, additional inputs such as DVD-D, USB, etc.

Performance: We used the DisplayMate benchmarking software to gauge the LCD monitors' performance. This software uses images to test a monitor for criteria such as Point Shape and Visibility. For example, this test can be used to determine how accurately the LCD

can display a fine point, and whether or not it is able to retain the round shape of a dot.

In the colour and greyscale set of tests, we checked for colour reproduction, the level-shift problem, and the streaking or ghosting effect. We ran the 16-intensities and 64-level primary and secondary colour intensities tests to see how well the monitors could display shades.

In the miscellaneous tests within DisplayMate, we checked for screen uniformity. We also tested the Reverse Video Contrast. The Pixel Persistence Test was run to check for jerks or image blurring.

We used the Passmark Monitor Test to check for image persistence at the edges. This is a measure of the LCD's response time.

For the viewable text angle test, we used a protractor placed in-line with the LCD panel, and the screen was viewed from either side. A document containing different fonts was viewed from various angles, letting us determine the angles at which the text still looked sharp. Similarly, for the viewable movie angle test, we viewed a movie from various angles and determined the angle at which discolouration of image began.

Price: We took the price of each monitor into account before assigning it a final score.

How We Awarded

The features, performance and price scores were given relevant weightages, and an overall score out of 100 was calculated.

The product that scored the highest was adjudged the winner of the Digit Best Buy Gold award for its category, while the second highest got the Digit Best Buy Silver award.